in the French Alps for an intense month of lectures and joint research. The site was the same Les Houches summer school, on the same green hillside opposite Mont Blanc, at which nine years earlier (1963) I had been taught the intricacies of general relativity (Chapter 10). In 1963 I had been a student. Now, in 1972, I was supposed to be an expert. In the mornings we "experts" lectured to each other and the students about the discoveries we had made during the past five years and about our current struggles toward new insights. During most afternoons we continued our current struggles: Igor Novikov and I closeted ourselves in a small log cabin and struggled to discover the laws that govern gas as it accretes into black holes and emits X-rays (Chapter 8), while on couches in the school's lounge my students Bill Press and Saul Teukolsky sought ways to discover whether a spinning black hole is stable against small perturbations (Chapter 7), and fifty meters above me on the hillside, James Bardeen, Brandon Carter, and Stephen Hawking joined forces to try to deduce from Einstein's general relativity equations the full set of laws that govern the evolution of black holes. The setting was idyllic, the physics delicious.

By the end of the month, Bardeen, Carter, and Hawking had consolidated their insights into a set of *laws of black-hole mechanics* that bore an amazing resemblance to the laws of thermodynamics. Each black-hole law, in fact, turned out to be identical to a thermodynamics law, if one only replaced the phrase "horizon area" by "entropy," and the phrase "horizon surface gravity" by "temperature." (The surface gravity, roughly speaking, is the strength of gravity's pull as felt by somebody at rest just above the horizon.)

When Bekenstein (who was one of the fifty students at the school) saw this perfect fit between the two sets of laws, he became more convinced than ever that the horizon area *is* the hole's entropy. Bardeen, Carter, Hawking, I, and the other experts, by contrast, saw in this fit a firm proof that the horizon area *cannot be* the hole's entropy in disguise. If it were, then similarly the surface gravity would have to be the hole's temperature in disguise, and that temperature would not be zero. However, the laws of thermodynamics insist that any and every object with a nonzero temperature must emit radiation, at least a little bit (that is how the radiators that warm some homes work), and everybody knew that black holes cannot emit anything. Radiation can fall into a black hole, but none can ever come out.